Future Generation Computer Systems 129 (2022) 152–169

Contents lists available at ScienceDirect

Future Generation Computer Systems

journal homepage: www.elsevier.com/locate/fgcs

Explainable framework for Glaucoma diagnosis by image processing and convolutional neural network synergy: Analysis with doctor evaluation

Omer Deperlioglu [a], Utku Kose [b], Deepak Gupta [c,*], Ashish Khanna [c], Fabio Giampaolo [d], Giancarlo Fortino [e]

[a] Afyon Kocatepe University, Afyonkarahisar, Turkey
[b] Suleyman Demirel University, Isparta, Turkey
[c] Maharaja Agrasen Institute of Technology, Delhi, India
[d] University of Naples Federico II, Naples, Italy
[e] University of Calabria, Italy

A R T I C L E I N F O

Article history:
Received 18 April 2021
Received in revised form 14 November 2021
Accepted 22 November 2021
Available online 6 December 2021

Keywords:
Glaucoma
Convolutional neural network
Explainable artificial intelligence
Class activation mapping
Image processing

A B S T R A C T

Glaucoma causes blindness in long-time untreated cases. So, its early diagnosis is very important. Moving from that, there have been lots of Deep Learning oriented studies to diagnose Glaucoma from color fundus images. However, it is important to know about how humans can trust to black-box level Deep Learning models in their decision makings. In this study, a hybrid solution with image processing and deep learning was supported with the Explainable Artificial Intelligence (XAI), to ensure a trustworthy decision-making for Glaucoma diagnosis. In detail, image processing employing both histogram equalization (HE) and contrast-limited adaptive HE (CLAHE) was used to enhance colored fundus image-data. For the diagnosis, the enhanced image-data was used by an explainable convolutional neural network (CNN). The XAI was achieved via Class Activation Mapping (CAM) allowing heat map-based explanations for the image analysis done by the CNN. The performance of the hybrid solution was tested with the Drishti-GS, ORIGA$^{-\text{Light}}$ and HRF retinal image datasets through a total of twenty attempts for classification. Considering the performance evaluation, the highest mean values were found with the ORIGA$^{-\text{Light}}$ dataset (with accuracy: 93.5%, sensitivity/ recall: 97.7%, specificity: 92.6%, precision: 93.8%, F1-Score: 95.7%, and AUC: 95.1%). As the XAI contribution of this study was with also an analysis by humans, the CAM based XAI effect was evaluated by some doctors. The CAM based XAI showed the accuracy of 82.73%, which was acceptable among alternative XAI methods. Also, according to the manual diagnosis test done with the doctors, the detections by the CAM based XAI was not below 97% and the worst diagnosis detection was 90%. Eventually, the results for the XAI effect were positive as pointing that use of XAI for black-box deep learning improves trust level for doctors.

1. Introduction

Glaucoma is a serious disease as the second highest reason for blindness (the first one is Cataracts). It is diagnosed with the symptoms like damage over the optic nerve head, high-level eye pressure, and gradual vision loss. In 2010, approximately 60 million patients with Glaucoma were known. That rate is predicted to be reaching 80 million by 2020 [1,2]. However, if the Glaucoma can be diagnosed early, it is possible to prevent from the existence of blindness. Because of that, early diagnosis is extremely important for urgent treatment. In this way, it is possible to eliminate the blindness potential [3,4].

Today, doctors often use intra ocular pressure measurement technique for diagnosing Glaucoma. Thanks to these tests, the optic nerve head appearance is observed to confirm the true diagnosis. As that requires periodic observations, the diagnosis of Glaucoma is an expensive process. Here, the diagnosis reliability is depending on the doctor's knowledge and experience. However, the doctors nowadays have also the advantage of using common technological tools. Because of that, many works have been done to ensure automatic diagnosis of Glaucoma from color fundus images [5–7].

* Correspondence to: Maharaja Agrasen Institute of Technology, Department of Computer Science & Engineering, Sector-22, Rohini, Delhi, India.
E-mail addresses: deperlioglu@aku.edu.tr (O. Deperlioglu), utkukose@sdu.edu.tr (U. Kose), deepakgupta@mait.ac.in (D. Gupta), ashishkhanna@mait.ac.in (A. Khanna), fabio.giampaolo@unina.it (F. Giampaolo), giancarlo.fortino@unical.it (G. Fortino).

https://doi.org/10.1016/j.future.2021.11.018

As general, Glaucoma is classified in two types: Primary Open-Angle Glaucoma (POA-G) and Angle-Closure Glaucoma (AC-G). POA-G, which accounts for about 90% of Glaucoma patients, is known as the most common type [8]. At this point, diagnosis of Glaucoma via retina image is a detailed procedure in which the extraction of various features from the data is required. In research studies, the retinal optic nerve head cup-to-disk ratio is considered as an essential indicator to detect the disease and the degree of the Glaucoma optic neuropathy. So, researchers have been mostly focused on automatic segmentation of the optical disc. On the other hand, the current research focus is related to correct detection of the optic vessel and the examination of the change. True way of segmentation for the optic vessel and the disk is very important in calculating the vessel/disk ratio and the diagnosis [9–13]. In this context, some recent research efforts are as follows: Khan et al. classified two features for the Glaucoma diagnosis with the retinal fundus image-data. The first feature was the cup-to-disk ratio and the second one was the neuro retinal rim ratio in the superior, inferior, nasal and temporal quadrants [14]. Yin et al. suggested that an effective enough computer-assisted system to diagnose Glaucoma can be achieved by automatic segmentation of the optic disc and the cup. In their study, they proposed a statistical method for segmenting the optic disc and the optic cup from the colored fundus image-data. That method introduced an optimal channel selection for information-based Circular Hough Transform and segmenting the optic disk [15]. Muramatsu et al. developed an automated tool for the detection of nerve fiber-layer defects in retinal fundus image-data. In this way, it was aimed to ensure an early detection of retinal nerve fiber-layer defects. Briefly, they tried to highlight the eye nerves by removing the vessels in the image [16]. On the other hand, Artificial Intelligence techniques have been effective to diagnose the Glaucoma disease from visual components such as optic coherence tomography (OCT) and fundus images. Artificial Intelligence allowed researchers to analyze or group the disease to determine its severity and progression. Thanks to the unstoppable advances of deep learning, a large number of potentials have been used in Ophthalmology, where traditional Artificial Intelligence solutions have been used for a long time. Specifically, deep learning techniques were applied via digital photographs, optical coherence tomography and other imaging techniques. These techniques have been used efficiently to evaluate the processes of many eye diseases such as Diabetic Retinopathy, Glaucoma, Cataracts and age-related Macular Degeneration [17–23]. In this context, the most important synergy of deep learning and image processing has been done via convolutional neural networks (CNN). As CNN models have been designed specifically for image processing, they are widely used within the processes of eye disease diagnosis from retinal image-data. Talking about the Glaucoma, some remarkable research efforts are as follows: Chen et al. proposed a deep learning solution for an automated Glaucoma diagnosis, which includes the removal of the corresponding optical disc region (ROI) and use of a CNN model. They briefly tried the enhance general performance of the proposed method by conducting extensive experiments in ORIGA as well as SCES dataset [18]. Raghavendra et al. proposed a CNN model applied over retinal fundus images, to ensure a computer-aided diagnosis of Glaucoma. They used an 18-layer CNN to improve performance in terms of classification through 1426 image-data including 589 normal and 837 Glaucoma cases (The achieved accuracy in this study was 98.13%) [24]. Norouzifard et al. developed a deep learning solution with VGG19 and Inception ResNet-V2 models. As a remarkable effort, this study trained the models with a very large dataset: ImageNet. The testing phase was done over different dataset like ONH dataset (so the generalizing abilities of the networks was shown). In terms of results, the VGG19 was not able to provide a generalizable framework, whereas Inception ResNet-V2 gave better accuracy values for validation, test and retest datasets [25].

1.1. XAI contribution and motivation

It is possible to replicate the past research efforts of Glaucoma diagnosis and derive alternative solutions. However, especially use of deep learning models means using black-box models. In such studies with black-box models, humans/doctors do not understand the limitations of the models and cannot answer the question of how they achieve good results in terms of diagnosis. So, it is important to support such diagnosis models with Explainable Artificial Intelligence (XAI) components, which are among hot Artificial Intelligence based solution components. XAI includes use of additional mathematical solutions to support interpretability of traditional machine learning models or making deep learning models explainable [26,27]. Here, especially image-data oriented models include use of additional explanation interfaces to show how the model reached to output data. Moving from that situation, the main objective of this study is to introduce a hybrid solution of image processing and Explainable Artificial Intelligence (XAI) supported deep learning, to ensure a trustworthy decision-making for Glaucoma diagnosis. As the contribution to the XAI literature, this study shows an alternative XAI view over a specific disease. In this way, it enables doctors to think again using automated intelligent systems for diagnosing from medical images. It is critical that especially automated deep learning models allow doctors to reduce time in terms of analysis over medical image data. But this study points that it is still possible to build trustworthy intelligent systems as XAI triggers deep learning models to reflect its decision-making process. Furthermore, there are not too many studies including XAI evaluations done by humans. Since XAI use in the context of medical field requires certain evaluations by doctors, this study employs a detailed evaluation phase including doctors' participation. So, this study contributes to the XAI literature by ensuring a bridge between human-side reflections about recent XAI use in deep learning. CNN is known as a popular deep learning model with its successful outcomes for medical applications. However, it still needs XAI components since it has many optimized parameters to show enough success level for target problems. As the target data is image, an effective XAI solution was done via CAM, which is an explainable interface for human view. In detail, that evaluation was done with even popular touch with hybrid formation over CNN and image processing techniques. It is thought that the outcomes of this study are valuable for not only researchers but also for doctors and medical staff. It is thought that the study will open eyes to see current potentials for trustworthy intelligent systems in disease diagnosis scenarios. It is also a nice motivation to follow that better deep learning models with additional efforts can be still supported with XAI to improve their interaction capabilities with human/user-side. Furthermore, this study follows the motivation that there is a great competition between intelligent systems and human diagnosing capacities but thanks to XAI solutions, there has to be no such dichotomy between datasets, computational tools (like deep learning and CNN employed in this study) and the skills developed by doctors' experiences. Moving from all explanations so far, the objectives of this study are based on all the mentioned motivations. It is also clear that the studies on the background do not meet with the requirements of effective use of XAI and deep learning for specific diseases and alternative interactions between the machine and human. So, this study has a critical potential for the associated research efforts.

Except from the XAI contributions, this study includes also efforts to design an uncomplicated method combining simple, widely used image processing techniques with deep learning-based. In detail, the image processing consisted of histogram equalization (HE) and the contrast-limited adaptive HE (CLAHE)

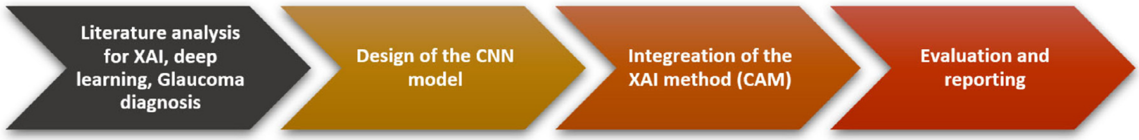


Fig. 1. General research flow of this study.

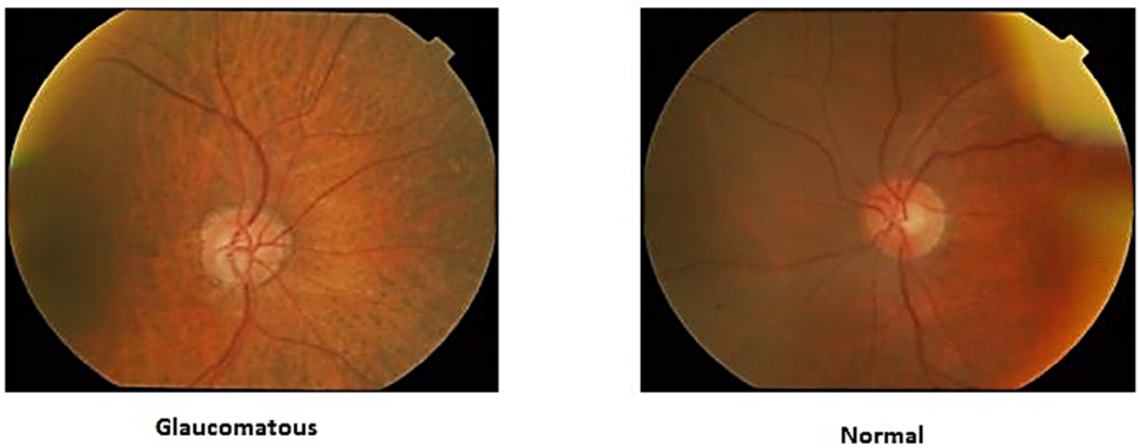


Fig. 2. Samples for glaucoma (drishtiGS_008.png) and normal (drishtiGS_004.png) cases.

to enhance colored fundus image-data, so that a processed input was used by an explainable convolutional neural network (CNN). Glaucoma diagnostics has been generally tried to be diagnosed by optic disc segmentation-based or features learning-based approaches. However, this study aimed to show that the diagnosis can be done efficiently, thanks to a practical hybrid system with traditional CNN and simple image processing techniques. The plain features of deep learning, which have been used recently to extract features, segment or classify, have also been revealed in this study. Furthermore, the trustworthy level of that hybrid system was evaluated by doctors so it was understood if XAI usage for black-box deep learning improves doctors' trust level against an automated Glaucoma diagnosis solution, which is rising over a hybrid approach.

Based on the main research point and motivations of this study, the overall plan of remaining content is as follows: The next section gives essential information regarding the employed components. After that, the third section explains details regarding the application side done with the explainable system. Next, the fourth section provides detailed information on the performed evaluation works and the obtained findings from them. Finally, the content is ended with discussions regarding conclusions and possible future work.

2. Materials and methods

In this study, the research was done by focusing on the associated literature first. After that, the next steps included designing the deep learning model (CNN), integrating the XAI method (CAM), performing evaluation works, and reporting the findings/results (Fig. 1). The rest of the paper provides detailed information associated with this flow.

Glaucoma affects the optic disc and optic nerves. It causes loses of retinal ganglion cells. At the same time, intraocular pressure increases. So, optic disc, optic nerve and order of blood vessels are all disrupt. If these issues are not treated, the further levels of the Glaucoma cause partial or complete blindness. Monitoring the optic disc and nerves to implement appropriate treatment is known among the best methods to delay advance of the disease and prevent from vision loss [28]. Here, the retinal images are photos showing the disease level and the progression according to the diagnosis of Glaucoma and the pathology of the retina. Fig. 2 shows samples for normal (drishtiGS_004.png) and Glaucoma (drishtiGS_008.png) cases.

In order to provide an alternative solution for Glaucoma diagnosis, this study consisted of two essential solution components: (1) image processing with HE, and CLAHE, and (2) the classification done via CNN. Furthermore, the CNN was supported with CAM based XAI to understand how the developed model establishes the bridge between input and the output. So, a black-box level hybrid solution was transformed into a trustworthy tool for doctors. That addition has been the main research motivation to contribute to the XAI literature.

Considering the whole solution process in this study, Fig. 3 represents a general flow chart towards the trustworthy Glaucoma diagnosis. Before evaluating the XAI effect, it is important to get a well-performing CNN model. In order to ensure that, Drishti-GS, ORIGA$^{-\text{Light}}$, and HRF retinal image datasets were employed for building the CNN model. In order to reduce the need for memory during the classification of the images and make a faster classification, image sizes in the dataset were decreased from original size of 1751 × 2049 pixels to 233 × 300 pixels, for the Drishti-GS. That decrease was from 1751 × 2049 pixels to 200 × 300 pixels for the ORIGA$^{-\text{Light}}$ dataset, and 2336 × 3504 pixels to 233 × 300 pixels for the HRF dataset. Following to that, the images were divided into R, G, and B parts. Next, the HE and CLAHE were applied to each part of the images separately. Finally, the R, G, and B parts were reassembled to obtain enhanced image versions. At the diagnosis side, the enhanced images were classified via 8-layer explainable CNN (with CAM integrated layer) to detect the cases and explain the decision-makings with heat maps over the input images.

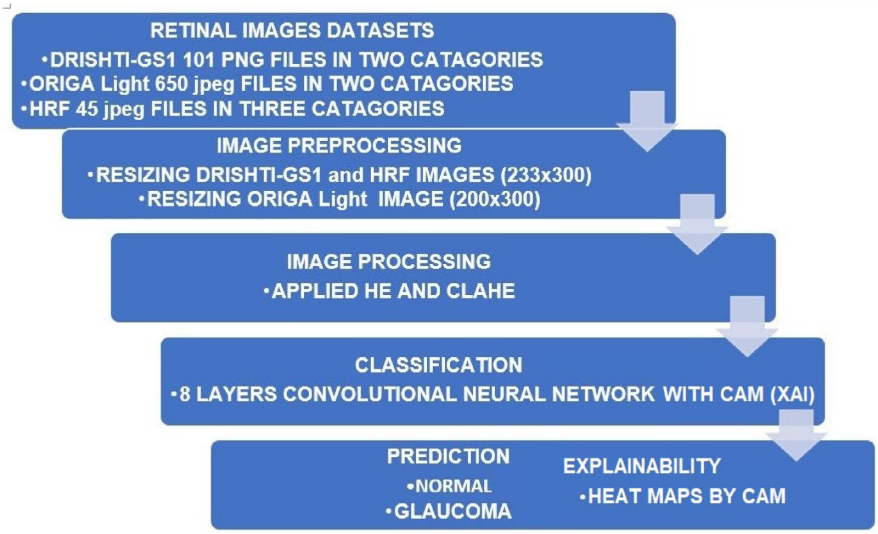


Fig. 3. Flow chart of the whole solution process in this study.

2.1. Retinal image datasets

In this study, Drishti-GS, ORIGA-Light and HRF retinal image datasets were used. Details regarding these datasets are as follows:

2.1.1. Drishti-GS: Retinal image dataset

The DRISHTI-GS dataset comes with a total of 101 retinal image-data, which are in two parts as 50 images are for the training, and the 51 images are for the testing phase. All images in the database were acquired by dilating the eyes with a 30-degree field-of-view on the optical disc and a 2896 × 1944-pixel size un-compressed PNG image. For each image, basic diagnoses were collected and marked by four Glaucoma experts with respectively 3, 5, 9 and 20 years of experience. In order to ensure a balanced data, low quality images with errors were removed from the dataset. Furthermore, the non-fundus black region around the fundus region was eliminated and the fundus images with 2049 × 1751 pixel-resolutions were obtained [29].

2.1.2. ORIGA^{-Light} retinal image dataset

The ORIGA^{-Light} dataset comes with retinal image-data collected during the SIMES: Singapore Malay-Eye Study, which was a community-based project for collecting retinal images. As funded by the National Medical Research Council, the project was run by the Singapore Eye Research Institute for a 3-year period (2004 to 2007). The SIMES examined a total of 3280 Malay adults, who are 40 to 80 years old, as 149 of them were diagnosed with Glaucoma positive. The retinal fundus image for both right and left eyes were collected by each subject. After that, each of them was examined by doctors. All image files were identified unique by removing any identifiable information. ORIGA-light dataset includes a total of 650 annotated retinal image-data in JPEG image format, with 168 Glaucoma and 482 normal cases. All image files are in the resolution of 2048 × 3072 pixels [12,30].

2.1.3. HRF: High-resolute fundus image dataset

The HRF dataset was created by a collaborative group in order to support studies of comparative evaluations for automated segmentation algorithms, which have been applied over retinal fundus image-data. As it is a public dataset, the HRF includes 15 images for normal people, 15 images for patients with Diabetic Retinopathy, and 15 images for the patients with Glaucoma. In terms of each image-data, binary gold standard vessel segmentation results are available as they were created by a group of experts and doctors [31].

2.2. Image processing phase

Before the diagnosis part, some image processing steps (Fig. 4) were applied over the related datasets. At the first step of the image processing, image sizes were adjusted/decreased from 1751 × 2049 to 233 × 300 pixels for the Drishti-GS dataset, 1751 × 2049 to 200 × 300 pixels for the ORIGA^{-Light} dataset, and 2336 × 3504 to 233 × 300 pixels for the HRF dataset. These changing resolutions and different resize outputs are because of changing details in different datasets. So, in order to ensure balanced input image-data for the CNN model, slightly different resize target pixels were determined. After that, the images were divided into R, G, and B parts. For each part, both HE and CLAHE were applied for part-oriented enhancements and eventually, all these R, G, B parts were reassembled to obtain the enhanced images, which will be input for the CNN model.

The image processing phase was important to enable the CNN model for processing images with enough details. At this point, both HE and CLAHE are simple and effective techniques according to more complex efforts like segmentation. In this context, essentials regarding both HE and CLAHE techniques are expressed briefly under the next paragraphs:

2.2.1. HE: Histogram equalization

The HE is among widely used image processing techniques for enhancement purposes. In detail, this technique makes target images enhanced to be seen more clearly. HE allows balancing the distribution of gray tones. That causes distinguishing the shades to be more clearly looking. In fact, the only process is to distribute the gray tones of the same pixel values to a larger scale. Thus, the difference between the color tones, which are close to others, are made lighter. So, the images are perceived better [32,33]. At this point, a conversion equation is required to perform this operation. In such equation, the average gray level should be found and every tone in the image should be proportioned to that average gray level so a general distribution should be obtained on the histogram view. That is done thanks to the following equation:

$$T_k = \sum_{j=0}^{k} \frac{n_j}{n} \tag{1}$$

In Eq. (1), the gray tone of each pixel in the old image is given as n_j and its equivalent in the new image is calculated accordingly. In case of colored images, R, G, and B parts are synchronized separately.

Fig. 4. The image processing phase for the target image-data.

2.2.2. CLAHE: Contrast-limited adaptive histogram equalization

Contrast-limited adaptive HE (CLAHE) is a technique especially for medical images as it has been showing good results for enhancing medical images. CLAHE is based on dividing image-data into nearly equal size, non-overlapping region parts. In order to obtain an effective statistical prediction, it divides 512 × 512 images into 8 regions of 64 pixels, by considering each direction. That partitioning process results in three different groups of regions. At this point, there are only four corners, and a group of these corners represents the class of corner zones. The second group of 24 regions represents the class of boarding regions. All regions belong to that class except the regions in the target image. The last group of the remaining 36 regions is defined as the class of the inner regions. In the CLAHE, the histogram of each region is calculated first. Then, a clip limit value is calculated in order to clip histograms by following a limit for the contrast enhancement. Each histogram is then rearranged so that the height value of a histogram does not exceed the clip boundary. Finally, the cumulative distribution functions of the histograms obtained for the grayscale mapping are determined. As the outcome of the CLAHE, the results obtained from the mappings regarding the four closest regions are linearly combined and mapped [34]. As it can be understood, CLAHE is an improved version of the HE, and it eliminates the disadvantage of the HE to increase noises in homogeneous regions of the image. That is done by the CLAHE by using several histograms targeting distinct section of the image, and distributing the lightness values to the image thanks to these histograms.

Following the image processing phase explained in the context of each technique and the general process, the next phase was diagnosis/classification of the Glaucoma, thanks to a XAI supported CNN model. The next sub-section is devoted to details regarding the model and the applications.

2.3. Classification of the glaucoma

The most critical part of this study was the classification since it includes results for both reasons: (1) diagnosis performance, (2) explainability/trust level for doctors (humans). Briefly, the enhanced images from the image processing phase were classified with a CNN model, which is supported by XAI.

2.3.1. CNN: Convolutional neural network model

CNN is a sub-type of discriminatory deep learning architecture achieving good performance in 2D data like images and videos with grid-like topology. In a typical CNN model, image convolution replaces the general matrix product seen in traditional neural network models. In this way, the total number of weights causes decreases in the complexity level of the whole network architecture. Also, since the image-data in the form of raw inputs can be transferred directly to the model, there is no need for additional feature extractions. Here, the CNN uses spatial relationships to have a smaller number of parameters. Eventually, general learning performance of such model is improved in terms of back propagation-oriented optimization methods. Also, CNN models have the advantage of requiring minimal levels of pretreatment [35].

Table 1
Details regarding the layers of the CNN model designed in this study.

Layer	Layer name	Details
1	Input layer	• Input jpg images • The image size of 256 × 300 × 1
2	Convolution layer	• The filter size: 4 • The number of filters: 16 (4 × 16)
3	ReLU layer	• A nonlinear activation function • The rectified linear unit function
4	Cross-channel normalization layer	• The size of the channel window: 2
5	Maximum pooling layer	• The size of the rectangle: [4, 3] • The step size function: Stride
6	Fully connected layer	• The number of classes in the target table: (2) • Weight Learn Rate Factor: 20 • Bias Learn Rate Factor: 20
7	Softmax layer	• The softmax activation function for classification
8	Classification layer	• The training parameters • The stochastic gradient descent with momentum • The maximum number of epochs: 100 • The initial learning rate: 0.00001

The CNN model employed in the application side of this study had 8 layers (details regarding these layers are given in Table 1). The inputs of the first layer are 2D, and the dimensions of the image are specified in the context of this layer [24]. In the study, 233 × 300 pixels is the defined image dimension and the value of 3 indicates that it receives an RGB image.

In CNN models, the convolution 2D layer runs convolution process over the input image. Here, the filter is created and transferred for the input image. After the convolution process, feature maps are obtained for the next layer. The dimensions of the network filter are shaped by the parameters of the convolutional layer. In the CNN model of this study, the filter size was 4. In addition, the parameter for the number of neurons (connected to the same output region), the number of filters, and the number of feature maps were all 16. The rectified linear unit (ReLu) is the sub-part of a typical convolutional layer where thresholding is performed for each element of the input. Thus, data redundancy is reduced and important features of the image are kept. The output of that layer has the same size with the previous layer, as pointed in [24]. In the study, designed CNN model used also a cross-channel normalization layer, as the channel window size was selected as 2 (for normalization). In CNN models, the maximum pooling layer is generally applied to each feature map to reduce the size. The maximum value on the filter is taken into account and the filter is replaced with that number. The output size for that layer is smaller than the output size of the previous layer. Moving from that, the CNN model in the study used the size of the rectangle as '4, 3'. Since the image is comprehensively

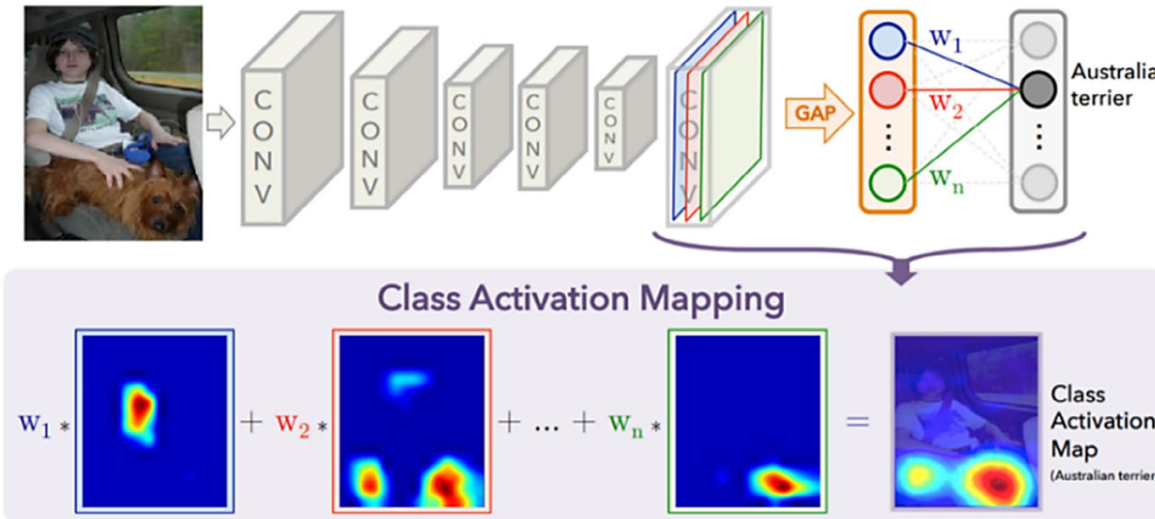


Fig. 5. CAM: Class Activation Mapping for ensuring explainability in CNN models [36,37].

scanned by the exercise function, the stride function was run for determining the stride size. By considering the neurons of the maximum pooling layer, a fully connected layer was employed for the classification step. The output of that layer is the number of classified classes. So, since the datasets of this study contained two output classes (as 'normal', and 'Glaucoma positive'), the output layer included two outputs. Additionally, the Weight Learn Rate Factor in the study was set as 20, and the Bias Learn Rate Factor was set as 20. It is critical to indicate that the Softmax layer of CNN models performs a softmax function on the input sample data. It is an exponential function that normalizes multidimensional data samples in the range of 0–1, which helps to reduce extreme values without removing them from the sample data. In addition, the classification layer contains the size, the output labels and the loss function [24]. In the CNN model of the study, the number of maximum epochs was set as 100. By using that designed CNN model and different training/test dataset groups, alternative classification applications were done for the Glaucoma diagnosis problem.

2.3.2. Explainable framework with CAM: Class activation mapping

For the explainable view of the designed CNN model, the Class Activation Mapping (CAM) was used in this study. CAM is a XAI method designed by the MIT-Computer Science and Artificial Intelligence Laboratory [36]. Fig. 5 represents a typical view for the mechanism of the CAM component. As it is seen from the figure, the CAM considers the weight values associated with the classification outcomes. In detail, the explainability is achieved by using Global Average Pooling (GAP) layers, which can diminish the image-data dimensions spatially so a decrease was done accordingly in terms of the parameters regarding feature maps [37]. The traditional CAM component comes with a score value in order to define linear combination of the pooled average feature maps. The GAP layers in this framework receive N channels to return spatial average value for each of them, then weight values are connected to each output data. Based on previously calculated parameters, a heat map is created for each class, by running the output image data from the previous layer towards calculation of a weighted sum as [37]:

$$w - 1 * f - 1 + w - 2 * f - 2 + \ldots + w - 2048 * f - 2048 \quad (2)$$

The resulting heat map is a simple interface for a human to understand which parts of the input image-data were considered by the CNN. That allows having clues for the decision through classification. In this way, the observation level of the CNN model was evaluated to detect any incorrect analysis over the input-data and track the successful detections via an explainable level [36,37].

2.4. Evaluation metrics

In this study, evaluations were done for both diagnosis and the explainability (XAI factor). After being sure about successful-enough diagnosis by the CNN model, the evaluations were done with active participants by some doctors. Just before explaining about applications and findings, some details about the evaluations are as follows:

2.4.1. Performance evaluation

In terms of deep learning applications, confusion matrix (truth table) is widely used for evaluating general classification performance. In order to evaluate the model well, it is essential to understand confusion matrix from various definitions of metrics [38]. As general, 7 different metrics are used to evaluate general performances in terms of diagnosis (classification) studies. These performance metrics are respectively accuracy, precision and specificity, sensitivity, recall, F-measure (F1-Score), and AUC. They are widely employed for calculating some values to reflect predictive ability and accuracy [39–42]:

$$\text{Accuracy} = \frac{tp + tn}{tp + fp + fn + tn} \quad (3)$$

$$\text{Sensitivity} = \frac{tp}{tp + fn} \quad (4)$$

$$\text{Specificity} = \frac{tn}{fp + tn} \quad (5)$$

$$Precision = \frac{tp}{tp + fp} \quad (6)$$

$$Recall = Sensitivity = \frac{tp}{tp + fn} \quad (7)$$

$$f\_measure = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

$$AUC = \frac{1}{2}\left(\frac{tp}{tp + fn} + \frac{tn}{tn + fp}\right) \quad (9)$$

In the related metrics/equations, tp corresponds to the number of real and true positives, fp is total number of false positive

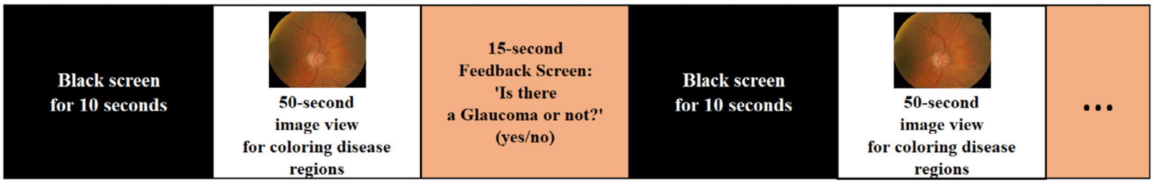


Fig. 6. General flow of the comparative test with manual diagnosis of Glaucoma.

diagnosis, *tn* is the real negative number, and finally, *fn* is the total false negative diagnosis.

By using the performance evaluation metrics, general state of the developed CNN model was understood in terms of its single state and its comparison with some competitive alternatives from the literature. So, a contributive perspective in terms of successful-enough diagnosis for Glaucoma was tried to be done first.

2.4.2. XAI: Explainable artificial intelligence evaluation

After being sure about the diagnosis performance, XAI ability of the designed CNN model was evaluated via some human-oriented evaluation works. In this context, a total of 15 doctors with the expertise on medical image-data and eye diseases participated actively in the followings (some characteristics regarding the doctors are shown in Table 2. Any personal, specific information is not provided to prevent from sharing any sensitive data):

- **Visual Evaluation:** Visualization by the CAM supported CNN model were examined to see if disease detection points were successfully observed by the black-box deep learning solution.
- **Usability Test:** In order to understand what doctors thought about interaction with the automated, explainable model, a usability test was run by employing several tasks (targeting usage of the diagnosis model) and gathering feedback in terms of task completion time, task completion rate, and task difficulty thoughts.
- **Comparing the CAM with Alternatives:** The evaluations regarding the XAI approach included a comparative work since there are alternative CAM methods introduced in the associated literature. Here, alternative CAM methods were integrated to the same CNN model and same applications (with same parameters, datasets…etc.) were run to get accuracy rates and feedback by doctors interacted with different XAI integrations.
- **Comparative Test with Manual Diagnosis:** By including all datasets, the 15 doctors were taken a specific manual diagnosis test where they used a coloring tool for indicating their Glaucoma diagnosis over each image-data. In the associated special test, the following three steps were done for each image-data: (1) viewing the image-data along 50 s, (2) viewing the feedback screen (asking if it is a Glaucoma case or not) running about 15 s, (3) a black screen viewed about 10 s to allow the doctor to refresh his/her mind for the next image-data. That test structure has been inspired from the neuroscience tests for focus, interest, and/or buying behaviors [34–36]. In order to eliminate the 'ordering effect' [37,38], the images were viewed randomly. At the end of that test, the doctors' feedback for each image-data was compared with the decision-makings by the CAM supported CNN. Fig. 6 shows the general flow of the test.

Table 2
Some characteristics regarding the doctors took part in this study.

Doctor No	Years of experience	Age	Gender	Previous experience with automated diagnosis
1	13	37	Male	Used machine learning solution before
2	21	64	Female	Used only screening software before
3	9	34	Male	Used only screening software before
4	17	40	Male	Used semi-automated diagnosis tool before
5	22	44	Female	Used machine learning solution before
6	16	39	Male	Used only screening software before
7	11	37	Female	Used only screening software before
8	14	37	Male	Used only screening software before
9	20	54	Female	Used semi-automated diagnosis tool before
10	11	33	Male	Used machine learning solution before
11	8	31	Male	Used machine learning solution before
12	9	35	Male	Used only screening software before
13	10	41	Female	Used semi-automated diagnosis tool before
14	10	44	Male	Used machine learning solution before
15	11	38	Male	Used semi-automated diagnosis tool before

3. Glaucoma diagnosis application

In the application side of the study, the image processing was performed first for all retinal images. In detail, the images were resized for faster classification. Then, the HE was applied to the image-data, which was separated into R, G, and B parts. Next, the CLAHE was applied as following the HE. CLAHE (*adapthisteq*) parameters were set as follows: number of tiles (NumTiles) as '3 3', contrast enhancement limit (clipLimit) as 0.01, range of output data (Range) as 'original', desired histogram shape (Distribution) as exponential, and distribution parameter (Alpha) as 0.1. After the use of CLAHE, the R, G, and B parts were reassembled to obtain the enhanced images. A sample of the obtained images through image processing steps is shown in Fig. 7.

One of the widely used measures for image enhancement is entropy. An image with enhanced features causes higher entropy values according to original version. In this study, the entropy of the original image for the drishtiGS_004.png was 6.9589 while it was 7.8633 for the enhanced version. The entropy of the original, and enhanced image for the drishtiGS_008.png was respectively 7.0192, and 7.8509. The entropy of the original image for the Im0499_g_ORIGA.jpg was 6.5631 while it was 7.7085 for the enhanced version. The entropy of the original image for the Im0001_ORIGA.jpg was 6.8696 and the entropy of the enhanced version was 7.7292. Considering such entropy increases among

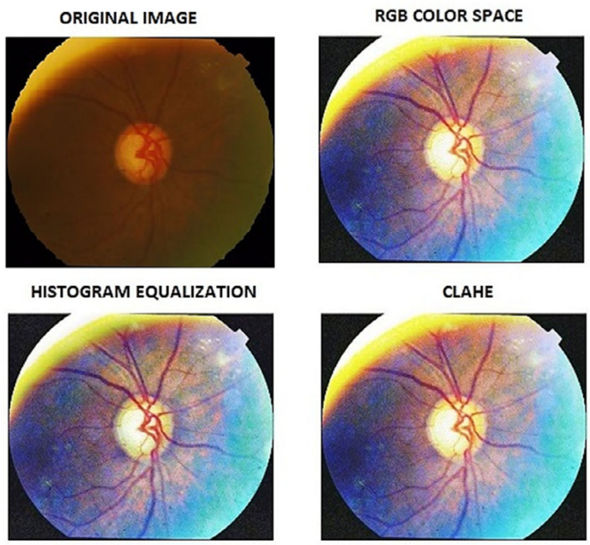


Fig. 7. Sample outcomes for each image processing step (drishtiGS_001.png).

all image-data and the obtained similar views like the ones represented in Fig. 7, it is possible to indicate that a good enhancement was seen in the images.

3.1. Performance evaluation findings

Classification applications were done to see the performance of the CNN model on diagnosing Glaucoma in three different datasets. Findings regarding each dataset are expressed in detail under the following paragraphs:

3.1.1. Findings for the Drishti-GS retinal image dataset

Classification applications were made with the training and test datasets of 3 different groups where randomly selected images from the Drishti-GS retinal image dataset were included. That was done in order to be sure if the performance by the designed CNN model was balanced among different data formations. Additionally, 20 different runs were done in all datasets. That was for seeing if the model was successful in independent runs for the same datasets. In the first group, 50% samples (50 images) were considered as the training dataset and 50% samples (51 images) were included in the test dataset. In the second group, 70% sample (71 images) were for training while 30% sample (30 images) were for the test dataset. In the third group, 80% samples (80 images) were for training dataset while 20% samples (21 images) formed the test dataset. The classification/diagnosis application was repeated a total of 20 times for each training and test data in three groups. The mean performance measures calculated after 20 trials of each group are given in Table 3, and visualized in Fig. 8.

As it can be seen in Table 3, and Fig. 8, the highest mean rates were obtained for the third data group with 80% training and 20% test data. That situation is quite normal for classification applications. Here, the question: 'so why that was necessary to be done?' comes to mind. Classification success rates were found very high in some applications in the literature. In these applications, majority of network trainings were done with the whole dataset. As the number of samples in the training dataset increases, it is aimed to show that the classification performance and the accuracy rate increases. The generalization feature of the network, which trained a large data, seems expanding.

Table 3
The mean performance findings for 20 trials in each three group regarding Drishti-GS.

Measure	Training 50%/ Testing 50%	Training 70%/ Testing 30%	Training 80%/ Testing 20%
Accuracy	81.20	87.10	91.10
Sensitivity	64.50	80.60	77.40
Specificity	84.90	91.30	90.70
Precision	71.40	78.10	92.30
Recall	64.50	80.60	77.40
F-measure	67.80	79.30	84.20
AUC	74.20	85.95	84.10

Table 4
The lowest, mean, and highest classification findings obtained in 20 trials for the third group.

Measure	Min (%)	Mean (%)	Max (%)	Std. Deviation
Accuracy	88.10	91.10	93.10	1.65
Sensitivity	64.50	77.40	77.40	1.54
Specificity	86.30	90.70	90.90	1.39
Precision	95.20	92.30	100.00	1.71
Recall	64.50	77.40	77.40	1.40
F-measure	76.90	84.20	87.30	1.45

The lowest and highest overall rates for the confusion matrices (obtained in 20 trials) for the third group are given in Fig. 9. For this application phase, the lowest overall (classification/diagnosis) accuracy value was 88.1% while the highest overall accuracy was 93.1%.

The lowest, mean, and highest overall classification mean performance measures obtained in 20 trials for the third group are given in Table 4, and visualized in Fig. 10.

As it is seen in Table 4, and Fig. 10, the mean value for the accuracy was 91.1% while the sensitivity/recall was 77.4%, specificity was 90.7%, precision was 92.3%, F1-Score (F-measure) was 84.2%, and the AUC was 84.1%, The findings with this study pointed that the developed image processing-deep learning synergy can work efficiently with different training/ test datasets. Furthermore, it also has the potential of achieving high forecasting capabilities. The obtained findings are generally close to or above the average values in previous studies. As general, the studies using segmentation, edge selection, and features extraction found classification accuracy rates between 73% and 96% for different retinal images datasets. In terms of all general studies, the accuracy values range from 90% to 94% [3,43]. In this study, the obtained results with essential image processing and deep learning were acceptable and above average values. As a comparison, some results from previous studies are given in Table 5. Better values are in green colors as the worst ones are in red and the medium values are in shades of yellow.

In Table 5, the best values are highlighted in bold style. At it is seen, the highest performance measures were found in [44]. However, Patil and Rao used the entire dataset (100 images) while training the CNN in their study. Because of that, the classification success seems high in [44]. Considering the designed explainable CNN model, the performance of diagnosing Glaucoma was good enough when compared with competitive studies.

3.1.2. Findings for the ORIGA^{-Light} retinal image dataset

The next classification applications were done with training and the test datasets from the ORIGA^{-Light} dataset. 80% samples (520 images) were chosen as the training dataset and 20% samples (130 images) were kept for the test. The diagnosis/classification process was repeated 20 times for different training/test data groups. The lowest, and highest overall value showing confusion matrices obtained in 20 trials are given in Fig. 11. In this context,

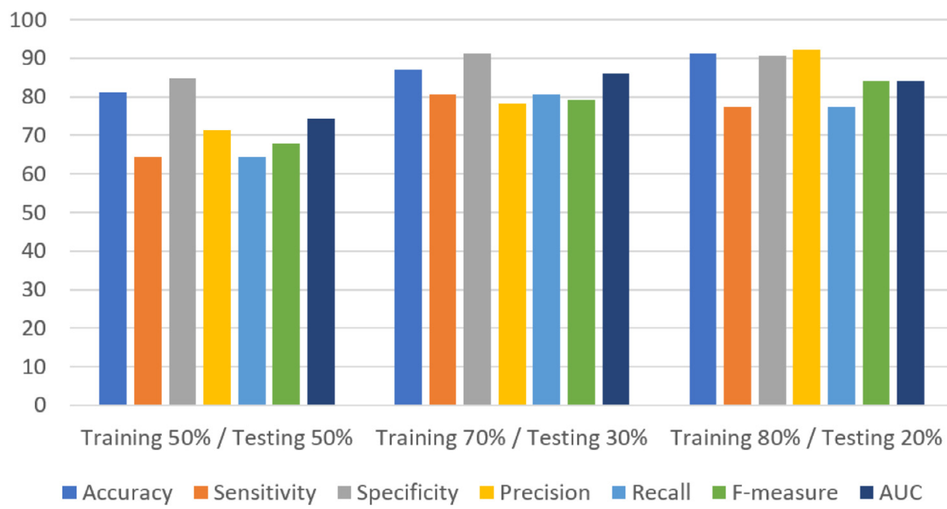


Fig. 8. Graphical view of the mean performance findings regarding Drishti-GS.

Fig. 9. The lowest and highest rates for the confusion matrices obtained in 20 trial-run for the third group.

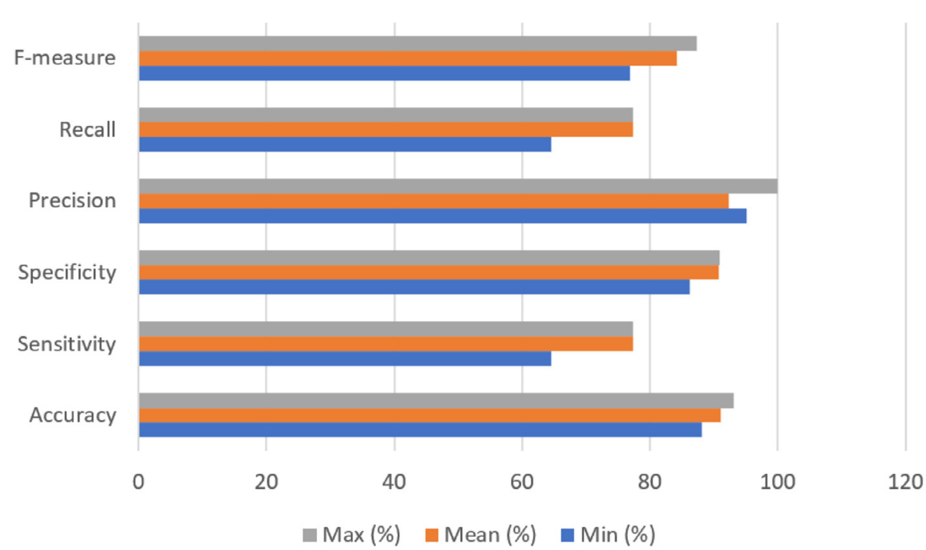


Fig. 10. Graphical view of the lowest, mean, and highest overall classification findings for the third group.

Table 5
Comparison of the obtained results with past studies done for the Drishti-GS retinal image dataset [43–51].

Model	Accuracy	Sensitivity	Specificity	Precision	Recall	F1-Score	AUC
This Study (XAI supported CNN)	91.10	77.40	90.70	92.30	77.40	84.20	84.10
U-Net [43]	92.10	80.00	62.00	-	-	-	-
Segmentation DL [44]	74.10	80.00	65.00	-	-	-	-
Stacked Auto-Encoder [51]	**98.20**	**99.60**	**94.60**	**94.60**	**99.60**	**97.90**	**94.00**
Multi-dimension DL [45]	72.80	80.30	85.50	-	-	-	90.30
DL with morphometric feature [46]	89.40	89.50	88.90	76.00	79.00	69.00	82.00
RNN [47]	88.70	87.20	90.00	-	-	-	-
Dense convolutional segment. [48]	88.20	85.00	90.80	-	-	-	-
CNN Transfer [49]	-	-	-	-	-	-	76.26
CNN [50]	75.25	74.19	71.43	-	-	-	80.41

Fig. 11. The lowest and highest rates for the confusion matrices obtained in 20-trial run for the Origa dataset.

Table 6
The lowest, mean, and highest classification findings obtained in 20 trials for the Origa dataset.

Measures	Min (%)	Mean (%)	Max (%)	Std. Deviation
Accuracy	92.90	93.50	94.00	1.51
Sensitivity	95.90	97.70	97.50	1.89
Specificity	84.70	92.60	92.20	1.70
Precision	94.70	93.80	94.60	1.81
Recall	95.90	97.70	97.50	1.21
F1-Score	94.90	95.70	96.00	1.18
AUC	90.30	95.10	94.80	1.42

the lowest overall accuracy value was 92.9% while the highest accuracy rate was 94.0%.

The lowest, mean, and highest overall classification mean performance measures (obtained in 20 trials) for the Origa^{Light} dataset are given in Table 6, and visualized in Fig. 12.

Both Table 6 and Fig. 12 point that the mean performance findings for the Origa dataset were higher than the ones for Drishti-GS dataset. Considering these, it was seen that an increase in the number of sample image-data (in the training and test data) and general quality of the images in the Origa^{Light} dataset affected the classification performance. In addition, it was seen that the generalization feature of the system selected positive trues better.

The mean overall performance measures obtained for the classification process of Origa^{Light} means that: The accuracy rate of 93.5% in the classification shows that the number of correct predictions made by the designed explainable CNN model is quite high. Recall or sensitivity rates show how many percent of patients with Glaucoma positive are correctly diagnosed by the model. The rate of 97.7% in the classification indicates that approximately almost 5 out of every 5 patients were correctly detected. The specificity ratio is an indicator telling the percentage of patients, who has no glaucoma, were correctly predicted by the model. The 92.6% ratio in the classification shows that there was a good solution approach by the model. The precision ratio in the classification shows that the 93.8% of the patients whom the proposed model diagnosed as Glaucoma positive are really Glaucoma positive (in the dataset), so the prediction mechanism of the model was shaped well. The rates of F1-Score 95.7%, and the AUC 95.1% indicate that the model distinguishes two different groups (Glaucoma and normal) well.

In order to have a comparative evaluation for the model performance, the findings obtained in previous studies were considered. The comparison of the obtained findings by previous studies for the Origa^{Light} Retinal Image Dataset are shown in Table 7. In the table, better values are in green colors as the worst ones are in red and the medium values are in shades of yellow.

Table 7 shows the highest values in bold style. As it can be understood from Table 6, the AUC metric was generally used to analyze the diagnosis/ classification performance in the whole studies employed the Origa^{Light} dataset. When AUC values are compared, it is seen that the highest value was obtained with the designed explainable CNN model. In fact, without use of clearly determined train/test sections, and without knowing the

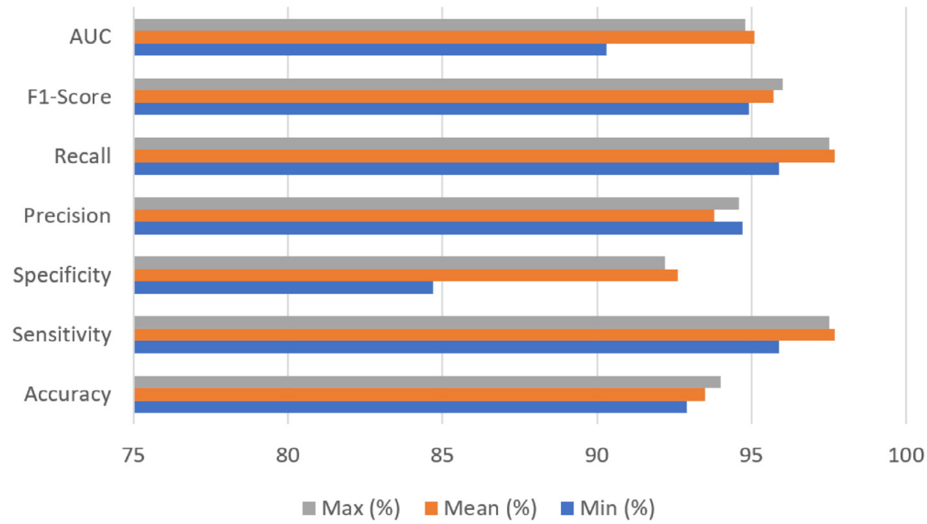


Fig. 12. Graphical view of the lowest, mean, and highest overall classification findings for the Origa dataset.

Table 7
The comparison of the obtained findings for Origa^{Light} retinal image dataset [14,49,52–55].

Model	Accuracy	Sensitivity	Specificity	Precision	Recall	F-measure	AUC
This Study (XAI supported CNN)	93.50	97.70	92.60	93.80	97.70	95.70	95.10
Automated segmentation [14]	-	-	-	-	-	-	83.10
Automated segmentation [14]	-	-	-	-	-	-	83.30
Dense conv. segment. [48]	99.86	87.68	99.94	-	87.68	-	74.43
Dense conv. segment. [48]	99.89	96.96	99.94	-	96.96	-	77.76
Two-stage localization DL [52]	79.67	79.38	71.17	77.97	79.38	77.88	87.40
Holistic, local deep features [53]	-	-	-	-	-	-	83.84
Holistic, local deep features [53]	-	-	-	-	-	-	71.87
Holistic, local deep features [53]	-	-	-	-	-	-	77.71
Holistic, local deep features [53]	-	-	-	-	-	-	77.81
Semi-supervised learning [54]	-	82.90	79.86	-	82.90	84.29	90.17
Deep residual learning [55]	-	77.75	80.55	-	77.75	81.37	86.07

distribution of normal and Glaucomatous image-data in both sets, AUC could not reflect the complete model state. At this point, alternative performance metrics such as accuracy, recall, and F-Scores should be seen in order to ensure a fair analysis and an objective comparison. However, in case of carefully-done train and test data formations, the AUC was sufficient to evaluate the classification mechanism of a model, as seen in [52]. Although the values found by Al-Bander et al. for accuracy, sensitivity–recall, and specificity values were higher, these were by the segmentation values corresponding to optic disc and optic curve. At this point, only the AUC was given for the classification. Therefore, the highest values were met with the designed CNN model in this study.

As it is known, both DRISHTI-GS and ORIGA^{Light} datasets contain Glaucoma class labels and OD/OC segmentations [56]. The obtained findings in this study indicate that the designed explainable CNN model can be run to classify retinal fundus images for diagnosis, without using segmentation. Because the CNN model of this study had better performance than the others with segmentation. Therefore, no time and effort were required to segment the optic disk or optic curve. Similarly, instead of increasing the classification cost by working with more complex CNN models (like VGG16, VGG19, ResNet50 etc.) effective results can be obtained with the synergy of image processing methods and an 8-layer explainable CNN model. It is remarkable that the determined parameters of the CLAHE have great importance in ensuring a simple image processing approach for image enhancement.

Table 8
The lowest, mean, and highest classification findings obtained in 20-trial run for the HRF dataset.

Measures	Min (%)	Mean (%)	Max (%)	Std. Deviation
Accuracy	86.70	88.90	91.10	1.37
Sensitivity	86.70	90.00	93.30	1.40
Specificity	83.30	85.80	88.20	1.89
Precision	87.70	87.10	87.50	1.31
Recall	86.70	90.00	93.30	1.27
F1-Score	87.20	88.50	90.30	1.47
AUC	85.00	87.90	90.80	1.56

3.1.3. Findings for the HRF retinal image dataset

The last diagnosis/classification application was done via HRF dataset. Here, 80% samples (30 images) were employed as the training dataset while 20% samples (15 images) were included in the testing phase. The diagnosis (classification) application was repeated 20 times for different training/test data groups. The lowest, and highest overall confusion matrices obtained in 20 trials for this dataset are given in Fig. 13. In this application, the lowest overall accuracy was 86.7% while the highest accuracy was 91.1%.

The lowest, mean, and highest overall classification mean performance findings obtained in 20-trial run for the HRF dataset are given in Table 8, and visualized in Fig. 14.

As Table 8 and Fig. 14 show, the mean values were respectively 88.9% for the accuracy, 90.0% for the sensitivity/recall, 85.8% for the specificity 87.1% for the precision, 88.5% for the F1-Score (F-measure), and 87.9% for the AUC. It is seen that the performance

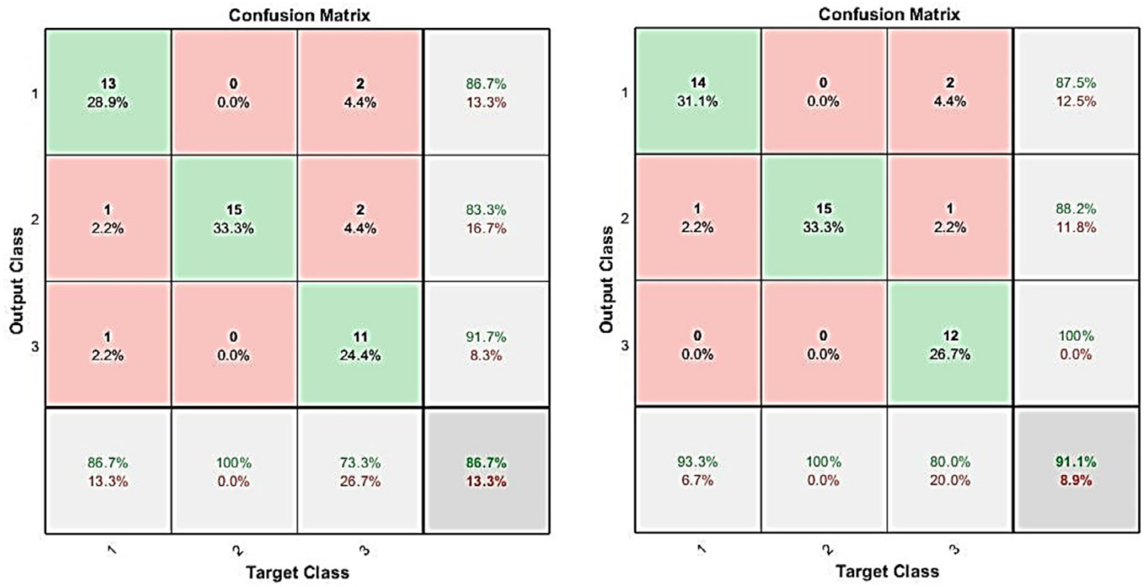


Fig. 13. The lowest, and highest rates for the confusion matrices obtained in 20 trials for the HRF dataset.

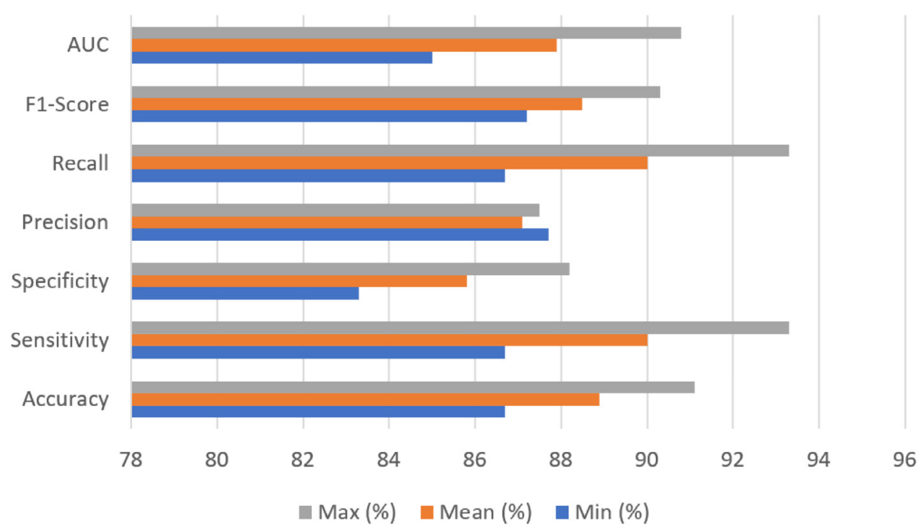


Fig. 14. Graphical view of the lowest, mean, and highest overall classification findings for the HRF dataset.

findings in this application are almost same with the findings obtained with the Drishti GS dataset.

In fact, more output classes caused lower classification accuracy rates in the studies with same neural network model [57]. Unlike the other two data sets in this study, there are three output classes in the HRF dataset: normal, Glaucoma and Diabetic Retinopathy. In addition, the number of sample images is low in the HRF dataset. Despite that, the designed explainable CNN model showed findings above average values. It is possible to indicate that the CNN model of this study was very effective in choosing the sample features. Also, although it was trained with few samples, the high rates for the prediction show that the designed model will be efficient in clinical applications and can be used not only for Glaucoma, but also for many other eye diseases such as Cataracts and Diabetic Retinopathy, and age-related Macular Degeneration.

3.2. XAI evaluation findings

As the CNN model was found to be effective enough in terms of Glaucoma diagnosis/classification, additional findings with the XAI evaluations were also considered. Findings regarding these evaluations are expressed under the following paragraphs.

3.2.1. Findings for the visual evaluation

Thanks to the CAM in the CNN model, the retinal fundus images were combined with the heat maps, in order to point which regions of each image were observed for the decision-making. Fig. 15 represents two resulting images considering different input image-data and their XAI-based analysis.

As it is seen from Fig. 15, intense colored heat maps are seen over the regions of non-normal vessel formations. That situation points the Glaucoma. Although a similar focus was seen in retinal image with no Glaucoma, it is critical that intense mappings are seen in Glaucoma – positive – images. That finding explains also that the designed CAM supported CNN model observed true regions like a human/doctor. Furthermore, in terms of technical manner, true parameters were employed to ensure bridge between input and output data, for the Glaucoma diagnosis. Of course, that way of evaluating a resulting heat map image may not be always a trustworthy indicator so 15 doctors took part in

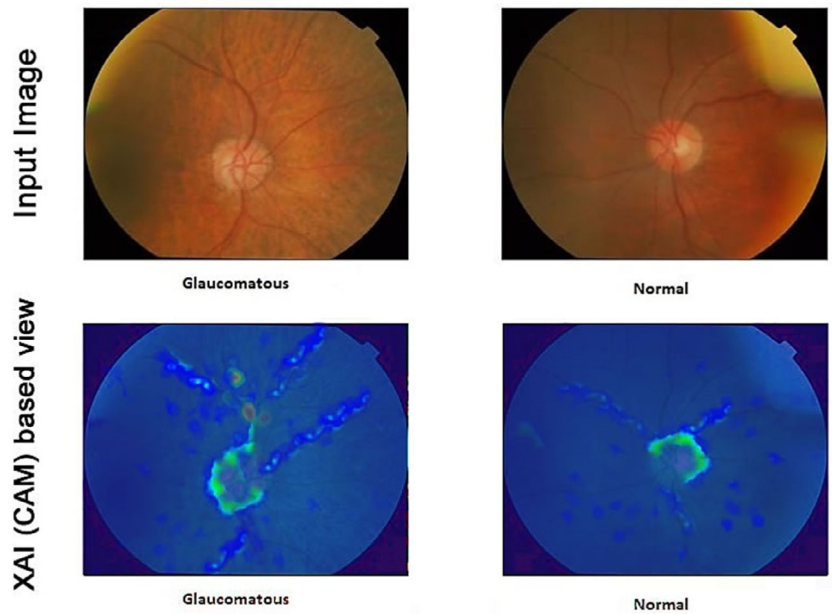


Fig. 15. CAM based outputs for the Glaucoma and normal diagnosis.

Table 9
Findings from the usability test of XAI based CNN model for Glaucoma diagnosis.

Task No	The Task	Completion Time (mean) / Top limit (seconds)	Completion Rate (%)	Difficulty (mean)
1	Run the automated diagnosis system.	3.43 / 5.00	100	1.32
2	Select a retinal fundus image from dataset.	6.82 / 9.00	100	1.18
3	Run CNN model for Glaucoma diagnosis.	16.44 / 25.00	100	1.68
4	Open diagnosis / classification findings.	6.09 / 8.00	100	1.36
5	Open / observe the XAI interface.	4.11 / 6.00	100	1.46
6	Reach to a decision for Glaucoma diagnosis.	3.49 / 5.00	100	1.10
7	Save / export the results.	4.32 / 5.00	100	1.07

The related findings are mean values by feedback of 15 doctors.

the following evaluation works targeting a detailed evaluation of the XAI solution.

3.2.2. Findings for the usability test

In order to have some findings regarding usability level of the XAI based CNN model, a usability test was performed with the active participation by 15 doctors. The doctors, who are experts in terms of eye diseases as well as image-data use, were formerly taken to a pre-training session. This session informed them about usage of the diagnosis model over the software environment. After the training session, the usability test was applied with some tasks to interact with the model (in terms of Glaucoma diagnosis) and observing the diagnosis findings by the CAM view. In detail, some parameters such as mean task completion time (in seconds), average completion rate (depending on the top completion limit for each task), and the mean difficulty feedback by the doctors (1: the task is easy, 2: the task is normal, 3: the task is difficult) were considered for a detailed analysis. Table 9 represents details and findings regarding the usability test. Better values in the Difficulty column are shown in green color while the worst values are in red and medium values are with white.

The findings of the usability test (Table 9) show that the doctors evaluated the software environment easy-to-use for general tasks on Glaucoma diagnosis. As it is seen from the findings about diagnosis and XAI views, the mean difficulty feedback in the associated tasks slightly increased. However, the findings were still within acceptable borders. In the test, the highest difficulty feedback values were respectively for task 3, and task 5. But these tasks were associated with main usage procedures as they did not reach to higher values. It is also notable that all tasks were completed with the rate of 100%, as based on the top limits determined for each task. Eventually, the interaction with the CNN model and the XAI (CAM) evaluated as positive by the doctors. Considering all tests, the mean completion time was 43.56 s. Table 10 also provides some additional findings regarding task completion time, easy-difficult tasks observed for each doctor.

When Tables 2 and 10 are matched, it is possible to see that age may be negative effect in dealing with such a computer-oriented solution. However, years of experience as well as past experience with automated diagnosis tools are positive factors to reduce total completion time. That may also affect some tasks, which are experienced easy or difficult (that may be a future work perspective on human–machine interaction with explainability support).

3.2.3. Findings for the compare of CAM with alternatives

At the time of this study was done, the CAM was not only XAI integration for the CNN-like models. There were several alternative XAI solutions designed for different data. Also, there were different variations of the 'default' CAM method. So, in addition to the CAM used mainly in this study, same CNN model was supported with alternative XAI variations as follows:

Table 10
Additional findings for each doctor's usability test experience.

Doctor No	Total completion time (s)	The easiest task completed[a]	The most difficult task completed[a]
1	38.10	2	7
2	44.40	1	4
3	35.16	2	4
4	39.41	1	4
5	39.55	2	4
6	29.57	3	7
7	33.48	1	4
8	37.06	1	7
9	42.47	1	3
10	28.39	2	4
11	31.28	1	4
12	33.41	2	3
13	36.29	3	4
14	34.29	1	3
15	30.19	2	7

[a]Indicated as task no. See Table 9 for task numbers.

- **Axiom-based Grad-CAM (A-G-CAM):** A-G-CAM (Axion-based Grad-CAM) was introduced as an enhanced visualization solution, which aims better performance than other CAM based solutions [58]. A-G-CAM follows the mechanism of eliminating disadvantages of the Grad-CAM, which is the second variation after the default CAM method.
- **Grad-CAM (G-CAM):** G-CAM (Grad-Cam) is a variation of the default CAM method, by employing gradients of the target output. Considering the last convolutional layer, the G-CAM ensures creation of a coarse localization map pointing the detected regions of the input image-data [59].
- **Attention Branch Network (ABN):** ABN (Attention Branch Network) is an alternative visual explanation method, which employs three components: feature extractor, attention branch, and perception branch. In this way, it allows combination of feature maps, attention solutions, and probability values to be used for creating explainable interfaces [60].

Considering alternative XAI methods, the same 8-layered CNN model was used by the doctors. For each alternative CNN model, the doctors were wanted to give feedback about usage and the XAI views. For the feedback, a total of four comparative statements were asked accordingly. Table 11 presents the findings by each doctor.

According to the findings in Table 11, the default CAM method was better in terms of stability. It was also found as an easily interpretable solution, by taking a common place among alternative methods. In terms of performance and view, ABN, and A-G-CAM integrations took slightly more top places according to the alternatives. However, there was not any certain borders among different CAM solutions and these findings might differ depending on CNN model formation as well as target diagnosis problem. Here, positive findings for the default CAM may be because it has a simple structure when it comes to integration. Also, a simpler mathematical background may be the reason for having a (more) homogenic visual view for better understanding level at the human side.

Table 12
Findings from the compare of the default CAM with alternative XAI methods in the CNN model.

Model	Accuracy (%)
Default CAM supported CNN model	82.73
A-G-CAM supported CNN model	83.26
G-CAM supported CNN model	80.61
ABN supported CNN model	87.23

In addition to the human side evaluation, the effect of changes for different XAI (CAM variations) integrations was analyzed in terms of accuracy (Table 12). Table 12 represents the best accuracy value with green color and the worst case with red color. As it can be seen from the table, the default CAM integration did not give a top performance (with slightly darker yellow color according to the A-G-CM supported CNN model) but it was still acceptable. For simple but effective trustworthy evaluations of black-box models, the default CAM seems like a good starting point to ensure enough interaction between a deep learning model and a human.

As a drawback of this study, it can be said that alternative XAI methods may be integrated to the same or similar CNN models. In this way, it may be seen if the current findings can be improved. Also, XAI contribution may be tested in different deep learning techniques and target medical image-data. It is thought that if more parameter catching XAI method can be used, the accuracy values of the XAI may be increased. However, a detailed evaluation with humans (doctors) should be still done. In this study, that was done as a unique contribution to the XAI literature. The details for that evaluation are expressed under the next paragraphs.

3.2.4. *Findings from compare with the manual diagnosis test*

Final evaluation work for the applied XAI integration was too critical because it allowed comparing of the detections by the CAM from each doctor's perspective. Each doctor took the manual diagnosis test explained before (Fig. 6). They evaluated 252 retinal fundus image-data (chosen as balanced from a total

Table 11
Doctors' feedback for the CAM and alternatives in the CNN model for Glaucoma diagnosis.

Doctor	Which XAI solutions(s) had the better performance in terms of running-time?	Which XAI solutions(s) had a better explainable view?	Which XAI solutions(s) did you find more stable?	Which XAI solutions(s) were easier to interpret via visual maps?
1	A-G-CAM	CAM	CAM/G-CAM	ABN/CAM
2	G-CAM	ABN/CAM	CAM	CAM/A-G-CAM
3	ABN	CAM	CAM/G-CAM	G-CAM
4	ABN/A-G-CAM	ABN/G-CAM/A-G-CAM	CAM/G-CAM	A-G-CAM
5	ABN	ABN/CAM	CAM	ABN/CAM
6	ABN/CAM	CAM/G-CAM	CAM/G-CAM	CAM/G-CAM
7	ABN/G-CAM	ABN/CAM	CAM	CAM
8	CAM	ABN/A-G-CAM	ABN	CAM/A-G-CAM
9	ABN/CAM	CAM	CAM	CAM
10	ABN	ABN/CAM	CAM	ABN/CAM
11	CAM/G-CAM	CAM	ABN/CAM	CAM
12	ABN/CAM	ABN/G-CAM/A-G-CAM	ABN/CAM	CAM/A-G-CAM
13	ABN/CAM	ABN/G-CAM/A-G-CAM	CAM	CAM/A-G-CAM
14	A-G-CAM	CAM/G-CAM	CAM	ABN/CAM
15	CAM/G-CAM	CAM/G-CAM	ABN/CAM	CAM

Table 13
Findings from the comparative evaluation with the manual diagnosis of Glaucoma.

Doctor	Successful Detection Rate by the CAM (%)			Glaucoma Region Detection Rate by CAM (%)	True Glaucoma Diagnosis Rate by CAM / CNN (%)
	Matching Rate '75%-85%'	Matching Rate '86%-95%'	Matching Rate '96%-100%'		
1	10	85	5	100	98
2	13	78	9	98	96
3	15	79	6	100	96
4	10	80	10	97	95
5	12	81	7	100	98
6	9	80	11	100	97
7	16	78	6	96	92
8	16	80	4	97	94
9	5	88	7	98	95
10	13	79	8	100	91
11	8	84	8	100	95
12	8	82	10	98	97
13	14	79	7	99	90
14	8	84	8	100	94
15	18	73	9	100	91

of 796 images considering all three datasets) within a total of 315-minute individual evaluation. In detail, the CAM based heat maps and human-side colorings were compared pixel by pixel. If the heat maps by CAM were above medium color levels and at least 75% of the regions matched with the regions colored by the doctor, the corresponding visual detection was labeled as successful. By considering successful detection of the CAM, the accuracy rate for the CNN was calculated, thanks to the matching levels: (1) 75%–85%, (2) 86%–95%, (3) 96%–100%. Additionally, true diagnosis (classification) rates of the CNN model were also calculated accordingly, by comparing with each doctor. Table 13 shows the findings from that comparative test. Better values for the detection rate columns are in green colors since the worst cases go towards red colors and medium values are with shades of yellow color.

When Table 13 is examined, it can be understood that the CAM method in the CNN model performed effective enough in detecting Glaucoma related regions in retinal images, so that diagnosing the disease. In detail, the visual matches were better in 86%–95% range so that the finding pointed good heat map colorings when compared with the human-side coloring. It was good to see that the region detection did not go down of 97% and also the worst diagnosis detection was 90%. Most of region detections were at 100% rate and the sign of at least 90% successful diagnosis also pointed that the CAM supported CNN model was trustworthy. Considering these, it can be said that such deep learning and XAI synergy can be used when doctors want to use a trustworthy, automated diagnosis model for Glaucoma. The findings here also show that there is a very low possibility for the CNN model to make errors in decision-making. On the other hand, it is another question that if it is possible to use traditional, interpretable machine learning systems for the same Glaucoma diagnosis problem. Based on the findings of this study and the outcomes for past deep learning-based studies, it can be said that use of interpretable machine learning will cause decreases in diagnosis performance. So, present and future medical diagnosis studies are greatly associated with use of deep learning and making them explainable via additional methods.

4. Conclusion

In this study, an explainable approach of image processing and deep learning synergy was introduced for Glaucoma diagnosis. The study aimed to contribute to the XAI literature by employing an explainable hybrid system, which was evaluated by doctors, to see if XAI is effective enough to eliminate trust anxieties regarding black-box models. Also, the study knows that there is a great competition between intelligent systems and human experts but by using XAI, it was pointed that there is no such dichotomy between datasets, computational tools and the skills developed by doctors' experiences. The overall plan of the paper was based on essential information regarding the employed components, details on the application side done with the explainable system, and reports as well as discussions on the findings from the evaluations. In order to ensure enhancement for the colored retina images, practical and widely-used technique formation including HE: histogram equalization, and CLAHE: contrast-limited adaptive HE was used accordingly. These techniques were implemented easily over medical image-data and the enhanced images were classified by an 8-layer CNN model. It is remarkable that the explainable side of the CNN was supported with the CAM: Class Activation Mapping to get heat map-based input data visualizations. By using the XAI, it was aimed to be sure about trustworthy level of that automated deep learning solution. The performance of the designed explainable CNN model was tested with three different retinal image datasets: Drishti-GS (with 101 retinal images), Origa^{Light} (with 650 retinal images) and HRF (with 45 retinal images) from the literature. The performance evaluations point the importance of image processing for effective and successful diagnosis of Glaucoma from enhanced retinal fundus images. In this way, deep learning was used without detailed edge selection or segmentation approaches, and an effective, explainable high-rate classification solution was obtained. Furthermore, the ability of the proposed model to detect Glaucoma was better than competitive studies. In addition to the performance evaluation, the most critical contribution of this study has been the XAI evaluations. These evaluations showed that a total of 15 doctors accepted the developed XAI-based solution as an effective and trustworthy diagnosis tool. Moving from the findings for the XAI perspective, it can be said that the integration of XAI improved human side trust level and usage interest for black-box deep learning models such as CNN. These findings also point a positive way for ensuring trustable, safe intelligent system formations for future medical applications. As the future seems associated with intense use of wearables for massive medical applications, the need for explainability will be met with such efforts introduced in this study. Depending on the data type, the CAM method and its alternatives are simple but effective tools for visual explanations.

Positive outcomes of this study motivated the authors for future works. In this context, alternative CNN models and CAM variations will be evaluated for the same Glaucoma datasets. Also, more studies will be done with alternative deep learning techniques and medical diagnosis problems (as meeting with the drawback/suggestion point indicated before). Furthermore, in order to contribute to the literature with alternative evaluations, more plans will be done for comparing deep learning and human-side performance states. Finally, another future work plan is associated with analyzing human-oriented factors (experience, age…etc.) with the XAI and deep learning usage.

CRediT authorship contribution statement

Omer Deperlioglu: Conceptualization, Methodology, Software, Supervision. **Utku Kose:** Data curation, Methodology, Writing – original draft. **Deepak Gupta:** Data curation, Investigation. **Ashish Khanna:** Visualization, Writing – original draft. **Fabio Giampaolo:** Software, Validation, Writing – review & editing. **Giancarlo Fortino:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] A. Almazroa, et al., Optic disc and optic cup segmentation methodologies for glaucoma image detection: a survey, J. Ophthalmol. (2015) http://dx.doi.org/10.1155/2015/180972.

[2] J. Lusthaus, I. Goldberg, Current management of glaucoma, Med. J. Aust. 210 (4) (2019) 180–187.

[3] H.A. Quigley, A.T. Broman, The number of people with glaucoma worldwide in 2010 and 2020, British J. Ophthal. 90 (3) (2006) 262–267.

[4] Y. Wu, M. Szymanska, Y. Hu, M.I. Fazal, N. Jiang, A.K. Yetisen, M.F. Cordeiro, Measures of disease activity in glaucoma, Biosens. Bioelectron. (2021) 113700.

[5] Muhammad Salman Haleem, et al., Automatic extraction of retinal features from colour retinal images for glaucoma diagnosis: a review, Comput. Med. Imaging Graph. 37 (7–8) (2013) 581–596.

[6] Ş.S. Kucur, G. Holló, R. Sznitman, A deep learning approach to automatic detection of early glaucoma from visual fields, PLoS One 13 (11) (2018) e0206081.

[7] J. Villamizar, G. Calderón, J. Carrillo, L. Bautista Rozo, J. Carrillo, J. Rueda, J. Castillo, Mimetic finite difference methods for restoration of fundus images for automatic detection of glaucoma suspects, Comput. Methods Biomech. Biom. Eng.: Imaging Visual. (2021) 1–8, http://dx.doi.org/10.1080/21681163.2021.1914733.

[8] Yandong Tang, et al., Automatic segmentation of the papilla in a fundus image based on the CV model and a shape restraint, in: 18th International Conference on Pattern Recognition (ICPR'06), Vol. 1, IEEE, 2006.

[9] J. Liu, et al., ARGALI: an automatic cup-to-disc ratio measurement system for glaucoma analysis using level-set image processing, in: 13th International Conference on Biomedical Engineering, Springer, Berlin, Heidelberg, 2009.

[10] D.W.K. Wong, et al., Automated Detection of Kinks from Blood Vessels for Optic Cup Segmentation in Retinal Images. Medical Imaging 2009: Computer-Aided Diagnosis, Vol. 7260, Int. Society for Optics and Photonics, 2009.

[11] Naoto Inoue, et al., Development of a simple diagnostic method for the glaucoma using ocular fundus pictures, in: 2005 IEEE Engineering in Medicine and Biology 27th Annual Conference, IEEE, 2006.

[12] Zhuo Zhang, et al., Origa^{-Light}: An online retinal fundus image database for glaucoma analysis and research, in: 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology, IEEE, 2010.

[13] Fauzia Khan, et al., Detection of glaucoma using retinal fundus images, in: The 6th 2013 Biomedical Engineering International Conference, IEEE, 2013.

[14] Fengshou Yin, et al., Automated segmentation of optic disc and optic cup in fundus images for glaucoma diagnosis, in: 2012 25th IEEE International Symposium on Computer-Based Medical Systems (CBMS), IEEE, 2012.

[15] Chisako Muramatsu, et al., Detection of retinal nerve fiber layer defects on retinal fundus images for early diagnosis of glaucoma, J. Biomed. Opt. 15.1 (2010) 016021.

[16] Cheng Zheng, jie, et al., Artificial intelligence in glaucoma, Curr. Opinion Ophthalmol. 30.2 (2019) 97–103.

[17] Parampal S. Grewal, et al., DL in ophthalmology: a review, Canad. J. Ophthal. 53.4 (2018) 309–313.

[18] X. Chen, Y. Xu, D.W.K. Wong, T.Y. Wong, J. Liu, Glaucoma detection based on deep convolutional neural network, in: 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2015, pp. 715–718.

[19] S. Qummar, F.G. Khan, S. Shah, A. Khan, S. Shamshirband, Z.U. Rehman, …, W. Jadoon, A deep learning ensemble approach for diabetic retinopathy detection, IEEE Access 7 (2019) 150530-150539.

[20] K. Shankar, A.R.W. Sait, D. Gupta, S.K. Lakshmanaprabu, A. Khanna, H.M. Pandey, Automated detection and classification of fundus diabetic retinopathy images using synergic deep learning model, Pattern Recognit. Lett. 133 (2020) 210–216.

[21] H. Zhang, K. Niu, Y. Xiong, W. Yang, Z. He, H. Song, Automatic cataract grading methods based on deep learning, Comput. Methods Programs Biomed. 182 (2019) 104978.

[22] C.S. Lee, D.M. Baughman, A.Y. Lee, Deep learning is effective for classifying normal versus age-related macular degeneration OCT images, Ophthalmol. Retina 1 (4) (2017) 322–327.

[23] J. Yim, R. Chopra, T. Spitz, J. Winkens, A. Obika, C. Kelly, …, J. De Fauw, Predicting conversion to wet age-related macular degeneration using deep learning, Nature Med. 26 (6) (2020) 892–899.

[24] U. Raghavendra, et al., Deep convolution neural network for accurate diagnosis of glaucoma using digital fundus images, Inform. Sci. 441 (2018) 41–49.

[25] Mohammad Norouzifard, et al., Automated glaucoma diagnosis using deep and transfer learning: Proposal of a system for clinical testing, in: 2018 Int. Conference on Image and Vision Computing New Zealand (IVCNZ), IEEE, 2018.

[26] A. Adadi, M. Berrada, Peeking inside the black-box: a survey on explainable artificial intelligence (XAI), IEEE Access 6 (2018) 52138–52160.

[27] A.B. Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, …, S. Tabik, A. Barbado, F. Herrera, Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI, Inf. Fusion 58 (2020) 82–115.

[28] F. Fumero, S. Alayón, J.L. Sanchez, J. Sigut, M. Gonzalez-Hernandez, RIM-ONE: An open retinal image database for optic nerve evaluation, in: 2011 24th International Symposium on Computer-Based Medical Systems (CBMS), IEEE, 2011, pp. 1–6.

[29] J. Sivaswamy, S.R. Krishnadas, G.D. Joshi, M. Jain, A.U.S. Tabish, Drishti-gs: Retinal image dataset for optic nerve head (onh) segmentation, in: 2014 IEEE 11th Int. Sym. on Biomedical Imaging (ISBI), IEEE, 2014, pp. 53–56.

[30] A. Cerentinia, D. Welfera, M.C. d'Ornellasa, C.J.P. Haygertb, G.N. Dottob, Automatic identification of glaucoma sing deep learning methods, in: Proc. 16th World Congr. Med. Health Informat. Precision Healthcare Through Informat.(MEDINFO), Vol. 245, 2018, p. 318.

[31] A. Budai, et al., Robust vessel segmentation in fundus images, Int. J. Biomed. Imaging (2013) http://www5.informatik.uni-erlangen.de/research/data/fundus-images/.

[32] O. Deperlioglu, U. Kose, Practical method for the underwater image enhancement with adjusted CLAHE, in: 2018 International Conference on Artificial Intelligence and Data Processing (IDAP), IEEE, 2018, pp. 1–6.

[33] D.J. Hemanth, O. Deperlioglu, U. Kose, An enhanced diabetic retinopathy detection and classification approach using deep convolutional neural network, Neural Comput. Appl. 32 (3) (2020) 707–721.

[34] Ali M. Reza, Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement, J. VLSI Signal Process. Syst. Signal, Image Video Technol. 38.1 (2004) 35–44.

[35] Francesco Piccialli, Vittorio Di Somma, Fabio Giampaolo, Salvatore Cuomo, Giancarlo Fortino, A survey on deep learning in medicine: why, how and when?, Information Fusion 66 (2021) 111–137.

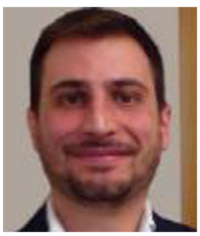
[36] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, A. Torralba, Learning deep features for discriminative localization, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 2921–2929.
[37] B. Gagana, Class activation maps. Medium blog (Gagana, B.), 2019, Online: https://medium.com/@GaganaB/class-activation-maps-551477720679, 2019 (Retrieved 3 2021).
[38] V. Kotu, B. Deshpande, Data Science: Concepts and Practice, Second edn., Morgan Kaufmann, Elsevier, 2019, pp. 263–279, http://dx.doi.org/10.1016/B978-0-12-814761-0.00008-3, 2019.
[39] Marina Sokolova, Guy Lapalme, A systematic analysis of performance measures for classification tasks, Inf. Process. Manage. 45.4 (2009) 427–437.
[40] O. Deperlioglu, Classification of segmented phonocardiograms by convolutional neural networks, BRAIN. Broad Res. Artif. Intell. Neurosci. 10.2 (2019) 5–13.
[41] O. Deperlioglu, Classification of phonocardiograms with convolutional neural networks, BRAIN. Broad Res. Artif. Intell. Neurosci. 9 (2) (2018) 23–33.
[42] A. Issac, M.P. Sarathi, M.K. Dutta, An adaptive threshold based image processing technique for improved glaucoma detection and classification, Comput. Methods Programs Biomed. 122 (2) (2015) 229–244.
[43] A. Sevastopolsky, Optic disc and cup segmentation methods for glaucoma detection with modification of U-Net convolutional neural network, Pattern Recognit. Image Anal. 27.3 (2017) 618–624.
[44] N. Patil, P.V. Rao, GlaucoNet: A highly robust stacked auto-encoder assisted deep learning model for glaucoma detection system, Int. J. Eng. Adv. Technol. (IJEAT) (5294) (2019) 9-1-5303.
[45] A. Chakravarty, J. Sivaswamy, Glaucoma Classification with a Fusion of Segmentation and Image-based Features, in: Proceedings of IEEE 13th International Symposium on Biomedical Imaging. 2016, pp. 13-16.
[46] J.Y. Choi, et al., Multi-categorical deep learning neural network to classify retinal images: a pilot study employing small database, PLoS One 12.11 (2017).
[47] O. Perdomo, et al., Glaucoma diagnosis from eye fundus images based on deep morphometric feature estimation, in: Comput. Pathol. Ophthalmic Med. Image Anal., Springer, Cham, 2018, pp. 319–327.
[48] P.P. Swapna, M.G. Mini, A regression neural network based glaucoma detection system using texture features, Int. J. Comput. Commun. Instrum. Eng. 3.2 (2016) 1–7.
[49] B. Al-Bander, B.M. Williams, W. Al-Nuaimy, M.A. Al-Taee, H. Pratt, Y. Zheng, Dense fully convolutional segmentation of the optic disc and cup in colour fundus for glaucoma diagnosis, Symmetry 10 (4) (2018) 87.
[50] J.I. Orlando, E. Prokofyeva, M. del Fresno, M.B. Blaschko, Convolutional neural network transfer for automated glaucoma identification, in: 12th International Symposium on Medical Information Processing and Analysis (Vol. 10160), International Society for Optics and Photonics, 2017, p. 101600U.
[51] A. Diaz-Pinto, S. Morales, V. Naranjo, T. Köhler, J.M. Mossi, A. Navea, Cnns for automatic glaucoma assessment using fundus images: an extensive validation, Biomed. Eng. Online 18 (1) (2019) 29.
[52] Muhammad Naseer Bajwa, et al., Two-stage framework for optic disc localization and glaucoma classification in retinal fundus images using deep learning, BMC Med. Inform. Decis. Making 19.1 (2019) 136.
[53] A. Li, J. Cheng, D.W.K. Wong, J. Liu, Integrating holistic and local deep features for glaucoma classification, in: 2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2016, pp. 1328–1331.
[54] A. Diaz-Pinto, A. Colomer, V. Naranjo, S. Morales, Y. Xu, A.F. Frangi, Retinal image synthesis and semi-supervised learning for glaucoma assessment, IEEE Trans. Med. Imaging 38 (9) (2019) 2211–2218.
[55] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016, pp. 770-778.
[56] J.I. Orlando, H. Fu, J.B. Breda, K. van Keer, D.R. Bathula, A. Diaz-Pinto, …, J. Lee, Refuge challenge: A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs, Med. Image Anal. 59 (2020) 101570.
[57] O. Deperlioglu, The effects of different training algorithms on the classification of medical databases using artificial neural networks, in: European Conference on Science, Art & Culture ECSAC, 2018.
[58] R. Fu, Q. Hu, X. Dong, Y. Guo, Y. Gao, B. Li, Axiom-based grad-cam: Towards accurate visualization and explanation of cnns, 2020, arXiv preprint arXiv:2008.02312.
[59] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE int. con. on comp. vision, 2017, pp. 618–626.
[60] H. Fukui, T. Hirakawa, T. Yamashita, H. Fujiyoshi, 2019. Attention branch network: Learning of attention mechanism for visual explanation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 10705–10714.

Dr. Omer Deperlioglu received his B.Sc. in Electric and Electronic (1988) Gazi University, M.Sc. in Computer Science (1996) from Afyon Kocatepe University, Ph.D. in Computer Science (2001) from Gazi University in Turkey. Now he is Professor of Computer Programming in Department of Science, Vocational School of Afyon, Afyon Kocatepe University of Afyon, Turkey. His current research interests include different aspects of Artificial Intelligence applied in Power Electronics, Biomedical, and Signal Processing. He has edited 3 books and (co-) authored 4 books and more than 30 papers, more than 30 conferences participation, member in International Technical Committee of 4 conferences and workshops. He also serves as one of the editors of the Biomedical and Robotics Healthcare book series by CRC Press.

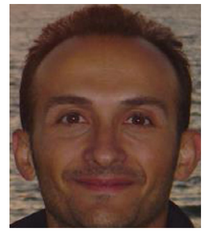
Dr. Utku Kose received the B.S. degree in 2008 from computer education of Gazi University, Turkey as a faculty valedictorian. He received M.S. degree in 2010 from Afyon Kocatepe University, Turkey in the field of computer and D.S./ Ph.D. degree in 2017 from Selcuk University, Turkey in the field of computer engineering. Between 2009 and 2011, he has worked as a Research Assistant in Afyon Kocatepe University. Following, he has also worked as a Lecturer and Vocational School — Vice Director in Afyon Kocatepe University between 2011 and 2012, as a Lecturer and Research Center Director in Usak University between 2012 and 2017, and as an Assistant Professor in Suleyman Demirel University between 2017 and 2019. Currently, he is an Associate Professor in Suleyman Demirel University, Turkey. He has more than 100 publications including articles, authored and edited books, proceedings, and reports. He is also in editorial boards of many scientific journals and serves as one of the editors of the Biomedical and Robotics Healthcare book series by CRC Press. His research interest includes artificial intelligence, machine ethics, artificial intelligence safety, optimization, the chaos theory, distance education, e-learning, computer education, and computer science.

Dr. Deepak Gupta (SMIEEE, MACM) received a B.Tech. degree in 2006 from the Guru Gobind Singh Indraprastha University, Delhi, India. He received M.E . degree in 2010 from Delhi Technological University, India and Ph.D. degree in 2017 from Dr. APJ Abdul Kalam Technical University (AKTU), Lucknow, India. He has completed his Post-Doc from National Institute of Telecommunications (Inatel), Brazil in 2018. He has co-authored more than 172 journal articles including 130 SCI papers and 51 conference articles. He has authored/edited 51 books, published by IEEE-Wiley, Elsevier, Springer, Wiley, CRC Press, DeGruyter and Katsons. He has filled four Indian patents. He is convener of ICICC, ICDAM & DoSCI Springer conferences series. Currently he is Associate Editor of Expert Systems (Wiley), and Intelligent Decision Technologies (IOS Press). Deepak Gupta have also been featured in the list of top 2% scientist/researcher database in the world [Table-S7-singleyr-2019]. He is also working towards promoting Startups and also serving as a Startup Consultant. He is also a series editor of ''Elsevier Biomedical Engineering'' at Academic Press, Elsevier, ''Intelligent Biomedical Data Analysis'' at De Gruyter, Germany, ''Explainable AI (XAI) for Engineering Applications'' at CRC Press. He is appointed as Consulting Editor at Elsevier.

Dr. Ashish Khanna (SMIEEE, MACM) received his Ph.D. degree from National Institute of Technology, Kurukshetra and his Post Doc (PDF) from Internet of Things Lab at Inatel, Brazil. He has around 140 research articles including 75 papers in SCI indexed Journals with cumulative impact factor of above 225 to his credit. He also has published 4 Patents. Additionally, He has authored, edited and editing more than 35 books for some of the top publishers like Springer, Wiley, Elsevier and many more. He has also served as a keynote speaker and resource person in several invited talks on national and international forums. He is also serving as Series Editor and consulting editor of top publishing houses like De Gruyter (Germany) Elsevier and CRC Press. He is currently working at the Department of Computer Science and Engineering, Maharaja Agrasen Institute of Technology, under GGSIPU, Delhi, India. He is convener of ICICC, ICDAM & DoSCI Springer conferences series. He is also working towards promoting Startups and also serving as a Startup Consultant. He has also worked for Smart India Hackathon. He has also completed a government funded research project.



Fabio Giampaolo is a research fellow in Computer Science and Mathematics at the Consorzio Nazionale Interuniversitario per l'Informatica (CINI). He received the Master Degree in Mathematics at the University of Naples Federico II. His research interests are focused on Machine Learning applications, Graph Mining and Data Visualization techniques.



Giancarlo Fortino (SM'12) is Full Professor of Computer Engineering at the Dept of Informatics, Modeling, Electronics, and Systems of the University of Calabria (Unical), Italy. He received a Ph.D. in Computer Engineering from Unical in 2000. He is also distinguished professor at Wuhan University of Technology and Huazhong Agricultural University (China), high-end expert at HUST (China), senior research fellow at the Italian ICAR-CNR Institute, CAS PIFI visiting scientist at SIAT — Shenzhen, and Distinguished Lecturer for IEEE Sensors Council. He is Web of Science Highly Cited Researcher 2020. He is the director of the SPEME lab at Unical as well as co-chair of Joint labs on IoT established between Unical and WUT and SMU and HZAU Chinese universities, respectively. His research interests include agent-based computing, wireless (body) sensor networks, and IoT. He is author of 500+ papers in int'l journals, conferences and books. He is (founding) series editor of IEEE Press Book Series on Human–Machine Systems and EiC of Springer Internet of Things series and AE of many int'l journals such as IEEE TAC, IEEE THMS, IEEE IoTJ, IEEE SJ, IEEE SMCM, IEEE OJEMB, IEEE OJCS, Information Fusion, JNCA, EAAI, etc. He organized as chair many int'l workshops and conferences (100+), was involved in a huge number of int'l conferences/workshops (500+) as IPC member, is/was guest-editor of many special issues (60+). He is cofounder and CEO of SenSysCal S.r.l., a Unical spinoff focused on innovative IoT systems. Fortino is currently member of the IEEE SMCS BoG and of the IEEE Press BoG, and chair of the IEEE SMCS Italian Chapter.